

Milestone 3: Differential Expression Analysis (DEA)

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Group number: G17

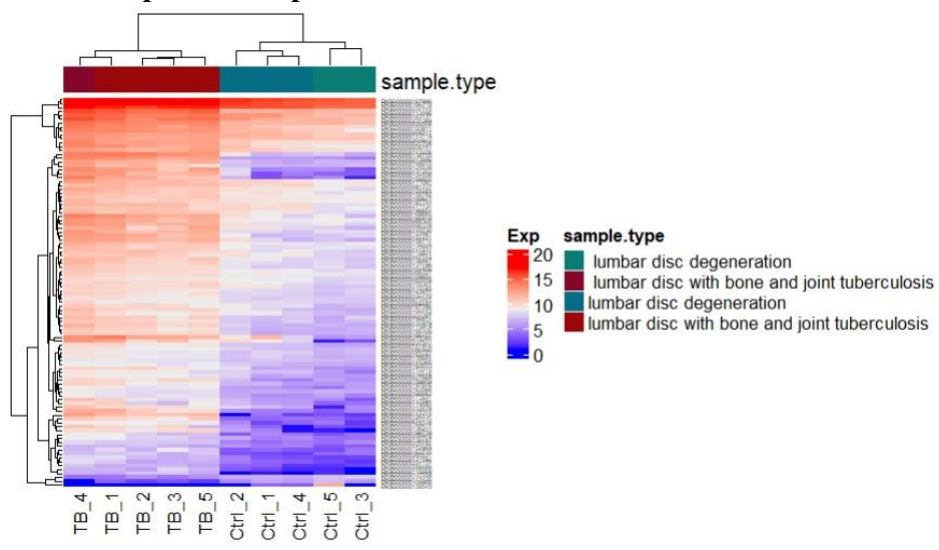
1. List of Differentially Expressed Genes (DEGs):

ENSG00000213886
ENSG00000074410
ENSG00000112096
ENSG00000169245
ENSG00000156234
ENSG00000101916
ENSG00000237541
ENSG00000025708
ENSG00000138755
ENSG00000115919
ENSG00000122861
ENSG00000117215
ENSG00000131203
ENSG00000005059
ENSG00000168685
ENSG00000186407
ENSG00000134028
ENSG00000081237
ENSG00000160326
ENSG00000275385
ENSG00000126545

(Full list is provided in the supplementary data).

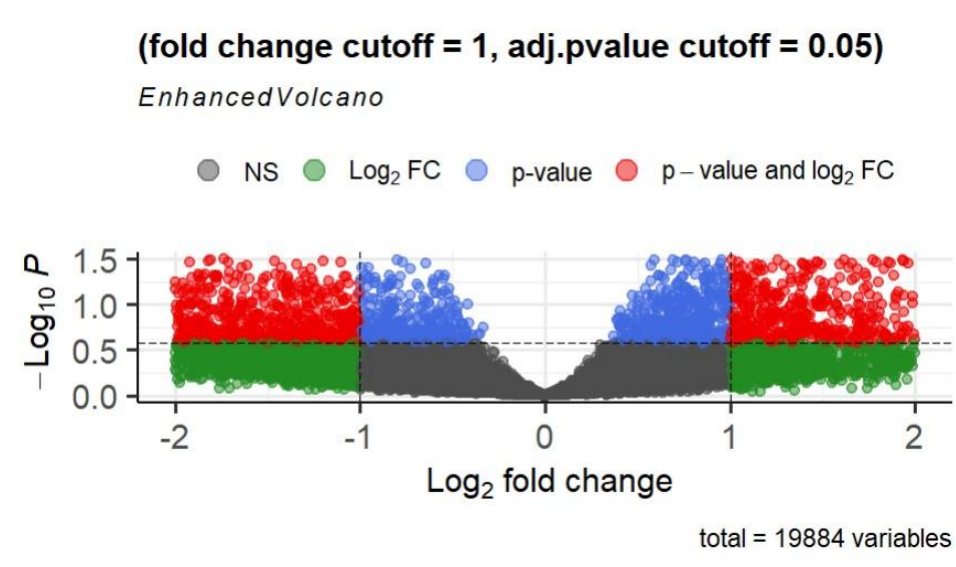
The following list represents the genes that showed significant differential expression between the control and disease groups. These genes were identified based on a statistical threshold p-value, log fold change (LFC) cutoff, and adjustive value (adj-pval). Identifying these DEGs is the primary step in understanding the molecular changes associated with the experimental conditions.

2. Heatmap of the Top 50 DEGs:



The heatmap represented the expression levels of the top 50 differentially expressed genes across all biological replicates. Our image indicates expression level: The color gradient (usually from blue to red). Red indicates up-regulation (higher expression), while blue indicates down-regulation (lower expression). The tree showed how genes and samples cluster together based on similarity in their expression profiles so that means higher expressed genes are lumbar discs with bone and joint tuberculosis and lower expressed is the lumbar disc degeneration.

3. Volcano Plot of the DEGs:



The Volcano plot is used to visualize the overall distribution of the differential expression genes results. In the middle area is log fold change have two regions, the upper is log fold change, the down is non-significant region based on LFC. The up regulated genes (on the right side) positive

and down regulated genes (on the left side) negative. The horizontal line means the p-value. The upper line significant ($p < 0.05$) and the lower line in non-significant.